Formula 1 Tyre Degradation Analysis

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ABSTRACT

In this analysis we aim to investigate tyre degradation in Formula 1 using data collected from the FastF1 API. Tyre degradation is one of the most important elements of Formula 1 strategy as tyre wear is one of the primary contributing factors to lap time. We will investigate this through the lens of fuel adjusted lap times, seeking to find what teams are best at managing tyres, the difference in degradation between compounds, and what tracks punish tyres the most. We created multiple different predictive models and analyzed trends to find answers to these questions and provide a framework for future improvements and analysis.

INTRODUCTION

There are many factors that go into F1 racing. Teams spend weeks analyzing data such as telemetry, driver, weather, fuel, pit stop, car components, and many other data points to inform their attempt at being competitive in a grand prix (race). For our analysis, we focused on the tyre degradation that occurs as a race goes on.

As a tyre degrades it loses its ability to grip the track, leading to slower lap times and shorter stints. There are many types of tyre degradation. Thermal degradation occurs when the rubber on the tyre overheats causing it to become greasy, ultimately reducing tyre grip. Abrasive wear occurs between the track and the tyre shaving off layers of the tyre. Blistering/Graining occurs when the tyre carcass is hotter than the cold surface or vice versa, leading to pieces of rubber exploding out of the tyre which reduces grip.

Different tracks will have different levels of degradation leading to a variation in tyre strategies from race to race. Some tracks have rougher surfaces and higher temperatures, which can lead to adding around 0.5 to over a second per lap during a race. As a tyre degrades, the driver will have to adjust and adapt their driving. If the rear tyres are more degraded than the front ones then they will feel oversteering, inducing wheelspin and causing sliding out on corner exits. If the front tyres are degraded, then they'll feel their car understeer. This leads to the driver driving more carefully as they adapt to the changes in their tyres, slowing down the overall time of the lap.

To address this issue, teams need to analyze their tyre degradation data to optimize strategies on how to manage their tyres. Figure 1 (next page) shows how different teams will switch out their tyres at different intervals in the race. These strategies change from race to race as teams account for the different conditions and tracks. They run simulations to create a plan going into the race but also adapt to the different situations and unexpected conditions that can occur during the race.

In this paper, we will explore the correlation between tyre life and lap time, examining how impactful tyre degradation is on lap times. Then we will be analyzing which teams are the best at managing their tyres, allowing them to pull ahead of the other teams. In the paper they analyze degradation between tyres but not for drivers and teams.

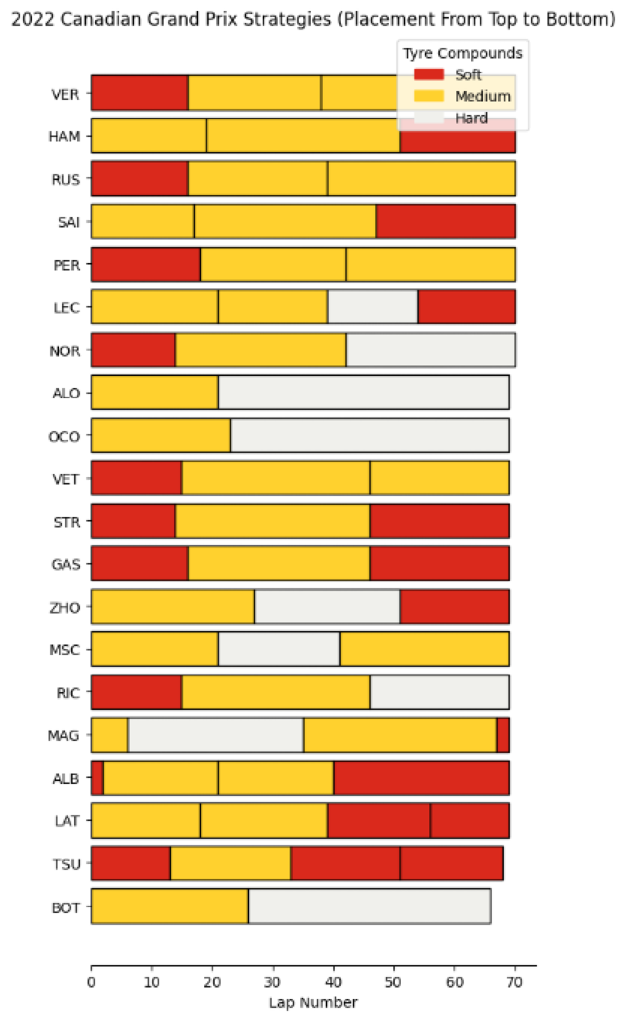


Figure 1

RELATED WORK

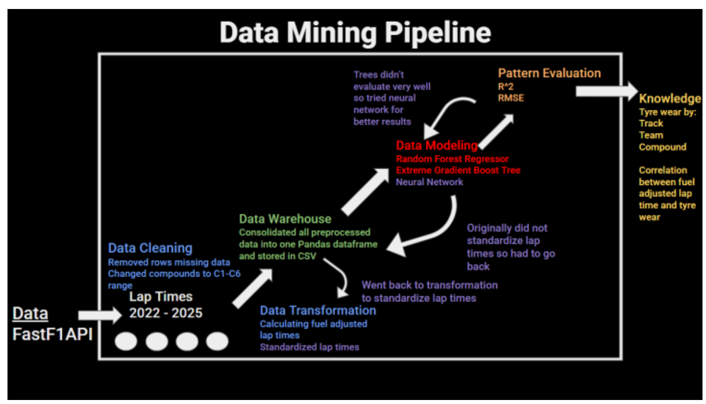
One work we referenced heavily was an article on fuel correction. The article explains why adjusting time for fuel correction is important, especially for the task of looking at tyre degradation. They then compare tyre strategies between drivers for one track, where we are looking into how teams manage tyre degradation and the correlations between tyre life and lap times.

Another similar paper "A State Space Approach to modeling Tire Degradation in Formula 1 Racing" by Cole Cappello and Andrew Hoegh. In this paper they explore using a Bayesian state-space model to estimate the degradation of tyres.

An article by catapult goes into how tyre degradation can be simulated using telemetry data. The article explains how degradation occurs, and how to minimize tyre degradation but doesn't go into details like finding the correlation between tyre degradation and lap time or comparing teams on their tyre management.

METHODOLOGY

We utilized Jupyter notebooks running Python to complete most of our data mining pipeline. Leveraging tools like Pandas, NumPy and Matplotlib gave us all the resources we needed to progress through the data mining pipeline as they are all very versatile tools that can be used for data modeling, analysis, visualization, and much more. Below is a chart of our data mining pipeline that we followed.



DATASET

We used the FastF1 API to collect our data. This API contains data back to the start of Formula 1 in 1950, however up until 2018 the amount of data collected and available to the public was very limited, normally only including lap times. This left us with roughly 900,000 rows of practice, qualifying, sprint race, and race data with each row signifying one lap completed by one driver. In 2022, there were regulation changes which means the 2018-2021 data and 2022-2025 data cannot really be compared.

From the 2022-2025 data we have around 300,000 laps from all sessions, however practice and qualifying data is often not representative due to a number of reasons. In practice, teams often following certain run plans which can involve alternating push and cool laps, run lower engine modes, and running unknown fuel amounts which makes this data impractical to use. For similar reasons qualifying data also cannot be used as these are laps done with minimal fuel in the car and often aim to use as much of the tyre as possible in a lap. This left us with just over 100,000 rows of race and sprint race data to analyze.

PREPROCESSING

The first preprocessing step was to change all the data of the tyre compounds to their respective compound type for each weekend. Pirelli brings three different tyre compounds from the range of the C1-C6 compounds to each race weekend based on several factors, and assigns a soft, medium, and hard tyre. In our dataset we are told if the drivers are on a soft, medium, or hard tyre however since these vary from track to track we had to manually look up the compound range for each weekend and utilize python functions to update the data for us.

The next step was to remove any rows where the “IsAccurate” label is set to false. This column was incredibly useful as it is true if the row is an accurate, representative lap time, and false if anything is wrong with the data. This includes missing data values, laps where drivers have come in/out of the pits for a tyre change, and safety cars as any of these will lead to the row not being good for analysis.

Fuel-Adjusted lap times

The final step of our preprocessing was to calculate the fuel adjusted lap times and calculate a percentage of lap time lost due to tyre wear for each row. Unfortunately, the weight and fuel data of cars is not public, so we are unable to calculate the precise amount of fuel burned per lap.

However, there are several things we know that can help us make a “good enough” estimate. The cars have a maximum fuel limit of 110 kilograms, and cars must also have at least one liter of fuel remaining in the car for sampling at the end of the race, so we can estimate they will consume about 109.25 kilograms of fuel (the density of the fuel is about 0.75kg/L). We can also assume that they uniformly burn this fuel throughout the race, as even though they may have some laps where they burn more or less fuel for whatever reason, teams are highly incentivized to burn as much fuel as possible to get more weight out of the car and more performance from the engine. We can then divide the amount of fuel burned evenly across every lap and calculate the approximate weight of fuel in the car.

The standard convention is that Formula 1 cars lose around three hundredths of a second per lap for each extra kilogram they have on the car. We can make a slightly more informed guess on the fuel effects based on if the circuit is a low, medium, or high-speed circuit. At lower speed circuits cars spend more time in the corners, and having more weight on the car makes the car less responsive and a little clumsier through the turns. This means at low-speed tracks we can assume the fuel effect is closer to four hundredths of a second per lap per kilogram, and alternatively at high-speed tracks where less time is spent in the corners, the fuel effect is closer to two hundredths.

Once we have the fuel for each lap and the fuel effect, we can then calculate the difference between the car at any fuel load and the car at a specific weight to remove the fuel effect from the car. For our project, we chose to balance the times at what the times would be if there was 55 kilograms of fuel in the car as this is roughly the average amount of fuel that would be in the car at any given time during the race distance.

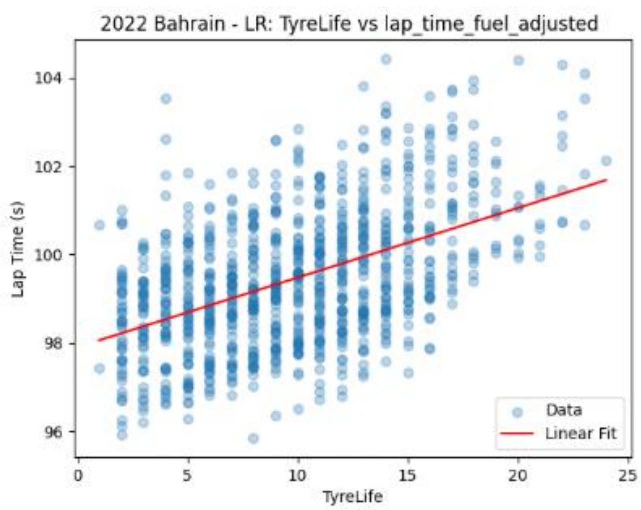
Finally, since lap times vary from track to track due to their characteristics, we must normalize these lap times somehow to compare data across tracks. The easiest way to do this is by comparing each lap in a weekend to the median lap time in the race and calculating the percentage difference. This gives us a percentage comparison to the median lap time for any given lap for any given race, which can then be used to compare tyre degradation data across tracks.

WAREHOUSING

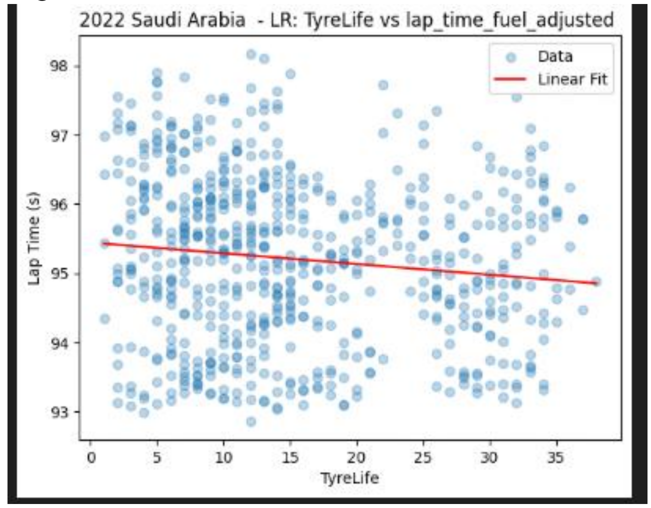
After preprocessing, we integrated all lap-level data from 2022-2025 from the FastF1 API into a single Pandas data frame. Each row is representative of one lap by one driver in one race, and includes data from the API along with our derived attributes such as fuel load, fuel standardized lap time, and normalized tyre degradation as a percent relative to the median lap time for a weekend. We then save this data frame to a CSV file which can be used for subsequent analysis and modelling.

MODELING

We created a couple of predictive models to find patterns between tyre life and lap times. First, we checked for a linear correlation between tyre life and lap time. At first we found a negative correlation where increases in tyre life led to lower lap times which didn't make sense. Then we realized that we needed to isolate the tyre life's effect on lap times. To do this, we did a couple things. First, we used the fuel-adjusted times as mentioned above. We then removed laps where the weather was rainy which could lead to inconsistent results. Finally, we removed lap times that were above 1.075 * median, since we didn't want laps that were due to driver error where the driver made mistakes to skew our results.

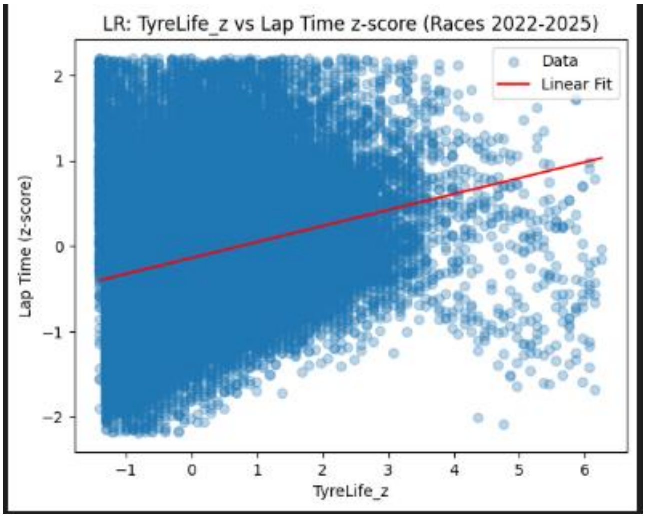


The figure above shows that for the 2022 Bahrain race there was a strong linear correlation between lap times and tyre life. However, this was not indicative of all tracks. For example, if we look at the figure below for Saudi Arabia 2022, we can see a slight negative correlation.



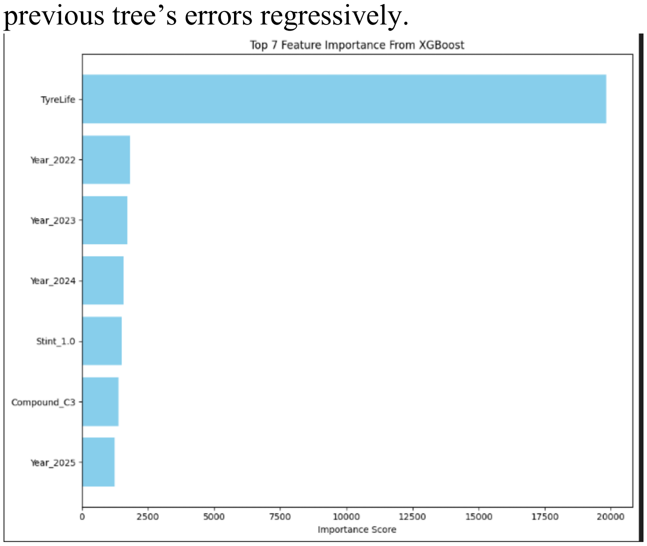
This is because some tracks benefit greatly from track evolution. Track evolution is a factor at every track, and involves rubber being laid down by cars as they circulate. As this rubber accumulates, the grip on the racing line is increased as the rubber on the circuit allows the rubber from the tyres to grip the track better. At a track like Saudi Arabia this effect is heightened as it is a purpose-built racing facility (meaning it is not converted for road use at any time) and it hardly gets any use, so the first laps are slow as the track is not rubbered in yet and leads to improvement in performance over time, even if the tyres are degrading.

Now to look at the correlation between tyre life and lap times across all races from 2022-2025, we need to standardize across the times. We did this by calculating the z score times for each lap within its race. Overall, we can see there is a very slight positive correlation; however, the R^2 score is 0.08, which is low as we expected and indicates that the relationship is not strictly linear.



To get better performance, we need to explore more models. We'll give the model more context for each lap, giving more features as input. The features we added on top of tyre life were: Year, Round, Driver, Team, Session, Tyre, Compound, Position, Stint. Since we have so many categorical features, a good step in the right direction is using decision trees.

Since we have so much data, we decided to use tree ensembles. This way we could see what was influencing our model's decision and hopefully be able to capture the non-linear parts of the correlation. The two models we tested were Random Forest Regressor and Extreme Gradient Boost Tree. The random forest regressor creates many independent trees and was a good model to get a baseline of R^2 of 0.57. The extreme gradient boost sequentially trains trees where each tree improves upon the previous tree's errors regressively.



After hyperparameter tuning it, this model has a R^2 of 0.668, which is not as high as we would like, but is much better than the linear regression and able to capture more patterns between tyre life and lap time. The figure above displays the weight of each feature on the prediction. As expected, tyre life has the biggest impact, followed by Year, Stint, and Compound. This is good since we wanted to find the relationship between tyre life and lap time, while many of the other features are mainly context that assists the model. Although some features' weight is not as high as the tyre life, they still impact the accuracy of the model. Without the year feature, the model has a large dip in R^2 down to 0.41.

The final model we used was a neural network. It was able to find more intricate patterns in the data. It gave us a R^2 of 0.87335, however since it is a neural network, it's difficult for us to see the intricacies and patterns it used to make predictions.

EVALUATION

	Random Forest Regressor	Extreme Gradiant Boost	Neural Network
R^2	0.57	0.668	0.871
RMSE	0.063	0.376	0.0546

Overall, the best model for predicting lap time using tyre wear was the neural network as it was able to find patterns in the data that the decision trees couldn't. We found that there is a meaningful correlation between tyre degradation and lap times. Our neural network model having a R^2 of 0.87 means the model can explain 87% of the variance in lap times. Although the relationship is not strictly linear, tyre we can conclude that tyre degradation has a strong influence on lap times.

DISCUSSION

There were numerous difficulties that we ran into throughout this project, all with varying natures and degrees of difficulty.

A lot of datasets that could give us more detailed information about the tyre degradation or fuel/weight specifically is either paywalled or private to the teams and drivers. This meant that we had to estimate the fuel on board along with the fuel effect as teams have ways of calculating fuel adjusted lap time with the data available to them.

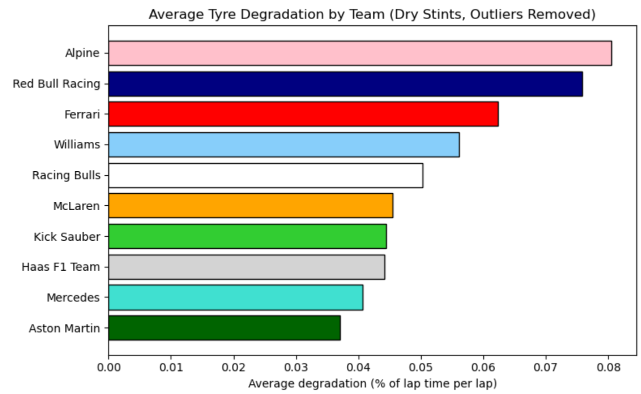
Another struggle was the scope of the data. Even though we had plenty of rounds with plenty of laps per round, we only got to look at four instances of a race around each circuit at most. This gives relatively limited stint data especially when looking driver by driver, so one outlier skews the data a lot.

Future work could include adding weather and temperature data to see how this further impact tyre degradation, either by adding this as an additional feature for the model or comparing these features to find a correlation. Further investigation could also be done into examining the differences between types of degradation (wear, thermal, graining, and blistering) and seeing how these different types of degradation affect lap times in their own unique ways. This could be done by looking at sector or sub-sector times to see where specifically time is lost across the lap or at telemetry data to see where errors are made. Different types of degradation lead to different issues in driving that manifest themselves more often in certain areas on track, so we could try and classify tracks on type of degradation using this analysis.

For me (Robyn), there was a background knowledge gap in my knowledge of F1 in general. And since we started the project late I didn't have enough time to learn more about the sport. This made it difficult for me to contextualize the data that I was looking at slowing down progress.

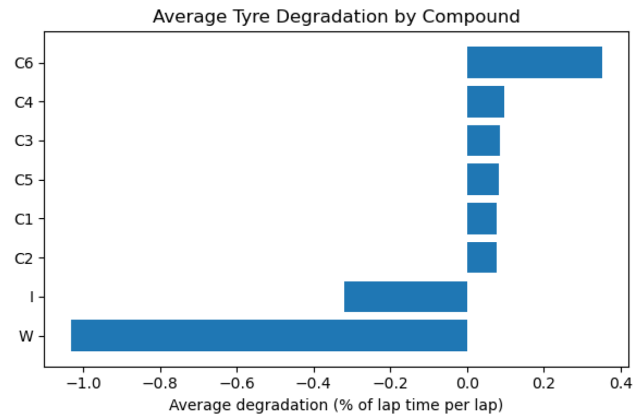
CONCLUSION

Looking into team performances, we found that Aston Martin has been the best across this regulation era at managing tyres as seen in the figure below, and Alpine has been the worst. These results are interesting as it shows that tyre degradation doesn't necessarily mean the car is worse as the winning driver of the drivers' championship in the last four years drives for Red Bull. This shows that while tyre degradation has a large impact on lap times, there are many other factors that go into performance and winning a race.

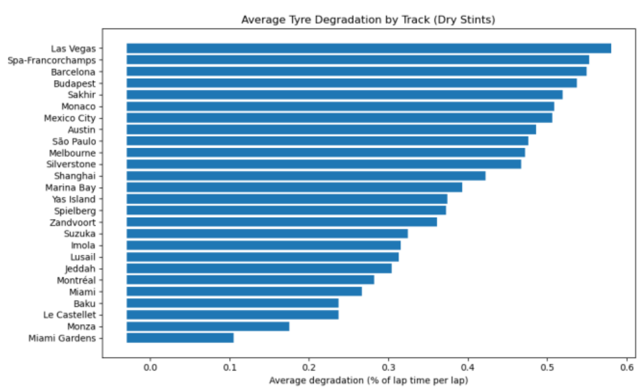


The data we found could be utilized by teams to influence strategies for future races to see what tracks they may have better or worse wear depending on their past trends. They could also see when other teams excel on tyre wear and potentially investigate, finding out what techniques or strategies they used to be so effective.

Looking at the average tyre degradation by compound, we can see a general trend for the harder tyres to wear less than the softer tyres, although not by much. The data on the C6 tyre is also minimal as it has only been used at a handful of grand prix since its introduction. Interestingly, we can see that the tyres reserved for rain races, the intermediate (I) and the wet (W) have negative degradation, however this is because the track is constantly improving when drivers are on these tyres because of the amount of water they dispel. These tyres do degrade; however, it is very difficult to calculate this since the conditions vary so much in wet conditions.



This data could also be used by Pirelli or the FIA to influence decisions on what tyre compounds they want to bring to each track. For weekends where Pirelli sees higher degradation, they may consider this for the next race run at the track and run a step harder tyres to ensure they are robust enough to survive the race distance. The FIA could also potentially use this information to influence Pirelli into bringing a softer range of tyres if they want to try and influence the teams into making more than one pit stop during the race distance.



A final application for this data would be for sports betting. General performance insights can be gained from this data including where drivers may be better on tyres than others, which may mean some weekends where tyre degradation comes more into play certain drivers may have advantages that aren't quite as evident on the surface. This data could also be used as a framework to predict the winning race margin, as the leading cars could be analyzed and the race margin estimated to a moderate degree of accuracy. Safety cars may come into play, but this could also be accounted for as some tracks have greater likelihood of safety cars than others.

REFERENCES

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Fuel correction: https://medium.com/@umakschually/fuel-correction-29ccd98ae62b

Tyre Compounds: https://www.pirelli.com/tires/en-us/motorsport/f1/tires

Tyre Degradation: **https://www.catapult.com/blog/how-tyre-degradation-affects-race-strategy**

Cole Cappello1 and Andrew Hoegh (2025): A State-Space Approach to Modeling Tire Degradation in Formula 1 Racing

APPENDIX

https://github.com/Robyn9999/F1-Analysis/tree/main

I, Robyn Luo, agree to the CU Boulder honor code.
Robyn Contributions:

- Initial and Checkpoint Project report/Presentation slide for Urban Planning Project
- Data gathering for Urban Planning Project
- Research on Urban Planning and models to use for geographical based data
- Data preprocessing/extraction for F1
- Data cleaning for F1 models
- F1 Model creation
- Final report/presentation for F1

Data Preprocessing
I, Aria Blondeau, agree to the CU Boulder honor code.
Aria Contributions:

- Initial and checkpoint project presentation/report for Urban Planning project
- Data preprocessing for Urban Planning project
- Project outline planning for F1 project
- Data transformation for F1 project
- Data warehousing for F1 project
- Evaluation/knowledge analysis for F1 project
- Final project presentation/report for F1 project